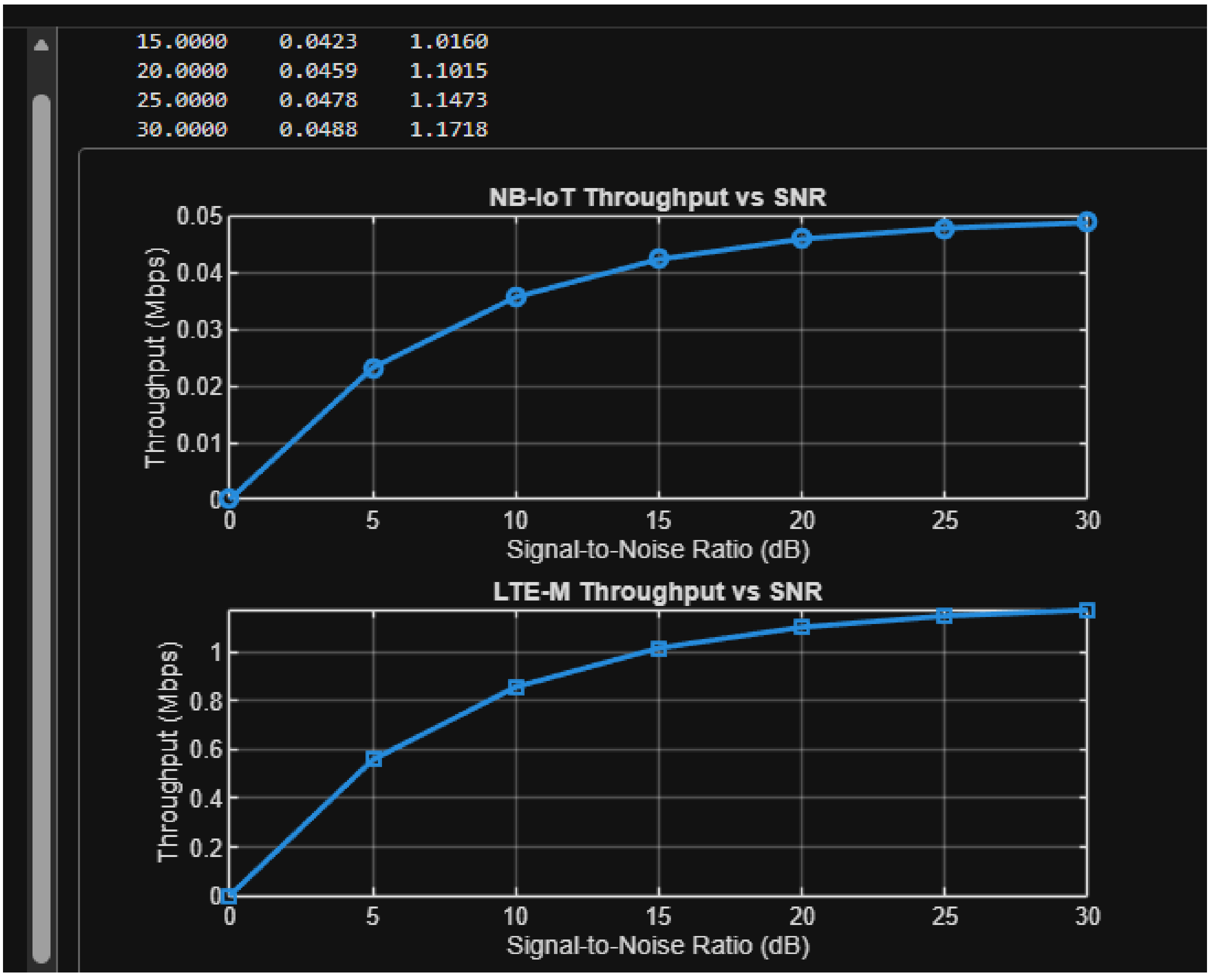
# Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation

## Objective 1 Matlab Code and Output:

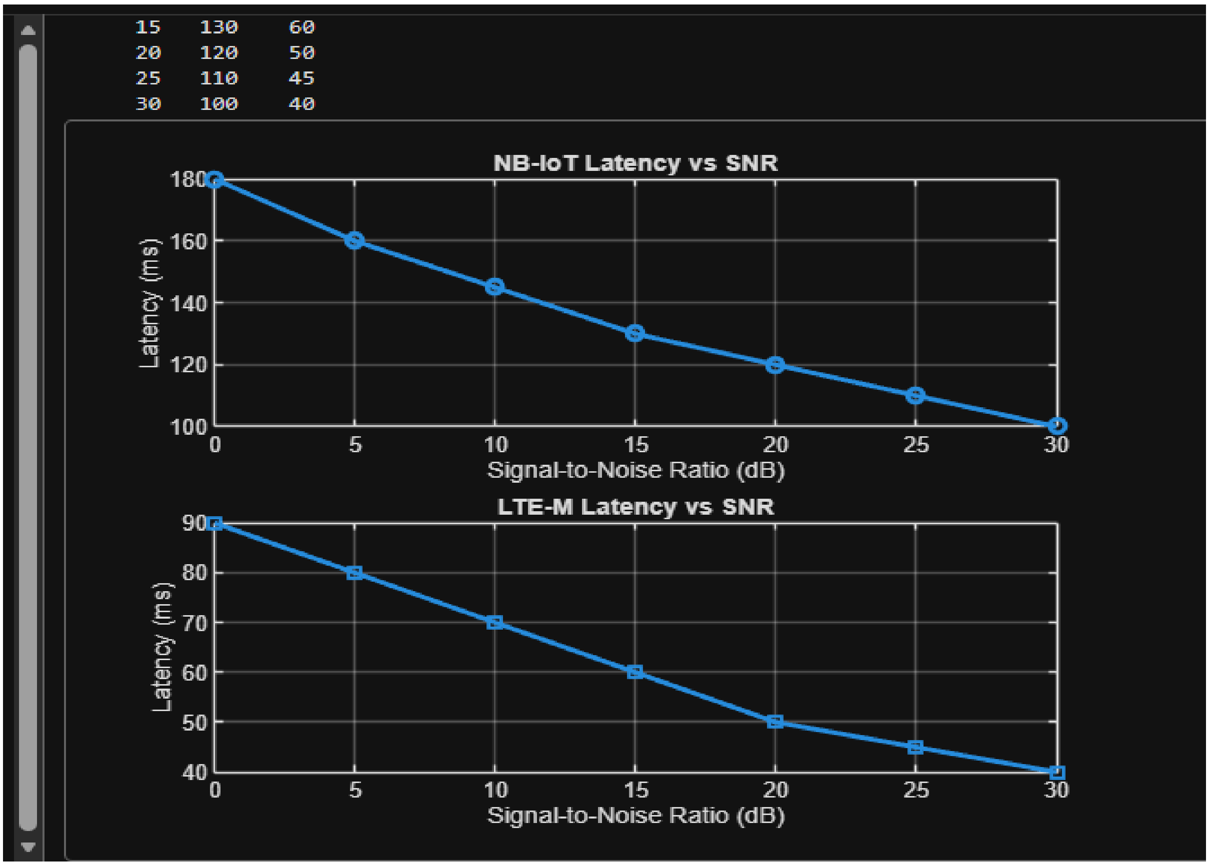
```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M  = 1.2*(1-exp(-SNR/8));
disp('SNR (dB)    NB-IoT(Mbps)    LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
```




## Objective 2 Matlab Code and Output:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100];

LTE_M  = [90 80 70 60 50 45 40];

disp('SNR(dB)   NB-IoT Latency(ms)   LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

figure
```

```
% -------- Graph 1 --------

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on
```



## Objective 3 Matlab Code and Output:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

NB_IoT = [82 86 89 92 95 97 99];

LTE_M  = [88 91 94 96 98 99 100];

disp('SNR(dB)   NB-IoT PDR(%)   LTE-M PDR(%)')

disp([SNR' NB_IoT' LTE_M'])

figure

% -------- Graph 1 --------

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on
```

| | | |
|---|---|---|
| 15 | 92 | 96 |
| 20 | 95 | 98 |
| 25 | 97 | 99 |
| 30 | 99 | 100 |